Suppose $f : M \to \mathbb{R}$ satisfies

$$\det\left(g_{j\bar{k}} + \partial_j \partial_{\bar{k}} f\right) = e^{F+f} \det(g_{j\bar{k}}) \qquad (*)$$

(g is metric with Kähler form ω), $\eta_{j\bar{k}} = g_{j\bar{k}} + \partial_j \partial_{\bar{k}} f$

$$\Rightarrow \quad Ric(\eta) = -\sqrt{-1}\,\partial\bar{\partial} F - \sqrt{-1}\,\partial\bar{\partial} f + Ric(\omega)$$
$$= -\omega - \sqrt{-1}\,\partial\bar{\partial} f = -\eta$$

So $Ric(\eta) = -\eta$

Continuity method can be used to solve $(*)$.

<u>Positive KE metrics</u>

PDE : $\det\left(g_{j\bar{k}} + \partial_j \partial_{\bar{k}} f\right) = e^{F-f} \det(g_{j\bar{k}})$

(ω Kähler form corresponding to g) $\quad Ric(\omega) - \omega = \sqrt{-1}\,\partial\bar{\partial} F$

<u>Continuity method</u>

$$(*_t) \quad \begin{cases} \det\left(\omega_{j\bar{k}} + \partial_j \partial_{\bar{k}} f_t\right) = e^{F - t f_t + c_t} \det(\omega_{j\bar{k}}) \\ \omega_{j\bar{k}} + \partial_j \partial_{\bar{k}} f_t > 0 \end{cases}$$

If $\omega_t = \omega + \sqrt{-1}\,\partial\bar{\partial} f_t$ then

$$Ric(\omega_t) = -\sqrt{-1}\,\partial\bar{\partial} F + t\sqrt{-1}\,\partial\bar{\partial} f_t + Ric(\omega)$$
$$= \omega + t\sqrt{-1}\,\partial\bar{\partial} f_t$$
$$Ric(\omega_t) = t\omega_t + (1-t)\omega$$

$(*_0)$ has solution by Yau's theorem
$(\mathrm{Ric}(\omega_0) = \omega)$

<u>Openness</u>: Let

$$\mathcal{F} : [0,1] \times C^{2,\alpha}(M) \times \mathbb{R} \longrightarrow C^{0,\alpha}(M)$$
$$(t, f, c) \longmapsto \log \frac{\det(\omega_{j\bar{k}} + \partial_j \partial_{\bar{k}} f)}{\det(\omega_{j\bar{k}})} - F + tf - c$$

linearization at (t, f_t, c_t):

$$D\mathcal{F}_{(t, f_t, c_t)}(\varphi, \lambda) = \Delta_{\omega_t} \varphi + t\varphi - \lambda$$

<u>Proposition</u> $L : C^{2,\alpha}(M) \times \mathbb{R} \longrightarrow C^{0,\alpha}(M)$
$$(\varphi, \lambda) \longmapsto \Delta_{\omega_t} \varphi + t\varphi - \lambda$$
is surjective.

<u>Proof</u> Key claim: if $\Delta_{\omega_t} \varphi + t\varphi = 0$ $(t \in [0,1])$
then φ is constant

We need to use that $\mathrm{Ric}(\omega_t) > t\omega_t$
so $\mathrm{Ric}(\omega_t) > (t+\varepsilon)\omega_t$ for some $\varepsilon > 0$

(all calculations are w.r.t. ω_t)

Suppose $\Delta_{\omega_t} \varphi + t\varphi = 0$

$$\Delta |\nabla \varphi|^2 = \nabla_p \nabla_{\bar{p}} (\nabla_j \varphi \, \nabla_{\bar{j}} \varphi)$$
$$= (\nabla_p \nabla_{\bar{p}} \nabla_j \varphi) \nabla_{\bar{j}} \varphi + \nabla_j \varphi \, \nabla_p \nabla_{\bar{p}} \nabla_{\bar{j}} \varphi$$
$$+ |\nabla_p \nabla_j \varphi|^2 + |\nabla_{\bar{p}} \nabla_j \varphi|^2$$

$$\nabla_p \nabla_{\bar{p}} \nabla_j \varphi = \nabla_p \nabla_j \nabla_{\bar{p}} \varphi = \nabla_j (\nabla_p \nabla_{\bar{p}} \varphi) = -t \nabla_j \varphi$$

$$\nabla_p \nabla_{\bar p} \nabla_{\bar j} \varphi = \nabla_p \nabla_{\bar j} \nabla_{\bar p} \varphi = \nabla_{\bar j}(\nabla_p \nabla_{\bar p} \varphi) + R_{i\bar p p \bar j} \nabla_{\bar i} \varphi$$

$$= -t \nabla_{\bar j} \varphi + R_{i\bar j} \nabla_{\bar i} \varphi .$$

So
$$\Delta |\nabla \varphi|^2 = -t|\nabla\varphi|^2 - t|\nabla\varphi|^2 + R_{i\bar j} \nabla_j \varphi \nabla_{\bar i} \varphi$$
$$+ |\nabla_p \nabla_j \varphi|^2 + |\nabla_{\bar p} \nabla_j \varphi|^2$$
$$\geq -2t|\nabla\varphi|^2 + (t+\varepsilon)|\nabla\varphi|^2$$
$$+ |\nabla_{\bar p} \nabla_j \varphi|^2$$
$$\geq -t|\nabla\varphi|^2 + \varepsilon|\nabla\varphi|^2 + |\nabla_{\bar p} \nabla_j \varphi|^2$$

$$0 = \int_M \Delta |\nabla\varphi|^2 \, dV \geq \int_M \Big(-t|\nabla\varphi|^2 + \varepsilon|\nabla\varphi|^2 + \underbrace{\nabla_{\bar p} \nabla_j \varphi \nabla_p \nabla_{\bar j} \varphi}_{t|\nabla\varphi|^2} \Big) dV$$

$$\int_M \nabla_{\bar p} \nabla_j \varphi \, \nabla_p \nabla_{\bar j} \varphi \, dV = -\int_M \nabla_j \varphi \, \nabla_{\bar p} \nabla_p \nabla_{\bar j} \varphi \, dV$$
$$= \int_M \nabla_{\bar j} \nabla_j \varphi \, \nabla_{\bar p} \nabla_p \varphi \, dV = \int_M (\Delta\varphi)^2 \, dV$$
$$= -t \int_M \varphi \Delta \varphi \, dV$$
$$= t \int_M |\nabla\varphi|^2 \, dV$$

$$\Rightarrow \quad 0 \geq \int_M \varepsilon |\nabla\varphi|^2 \, dV$$

so $|\nabla\varphi| = 0$, i.e. φ is constant. $\square$

Closedness is not true. There are no a priori estimates for $(*_t)$

Not a superficial difficulty:

Not all (M, ω) with $\omega \in 2\pi c_1(M)$ admit $\eta \in [\omega]$ s.t. $Ric(\eta) = \eta$.

<u>E.g.</u> blowup $Bl_p \mathbb{CP}^2$ does not admit such a KE metric.

<u>E.g.</u> Let $\omega = 2\omega_{FS}$ on $\mathbb{CP}^1$

In chart U_0 $(z_0 \neq 0)$ in terms of coordinate $x = \frac{z_1}{z_0}$

using section $s = (1, x)$:

$$\omega = 2\omega_{FS} = 2\sqrt{-1}\,\partial\bar\partial \log(1 + |x|^2)$$
$$= 2\sqrt{-1}\,\frac{dx \wedge d\bar{x}}{(1+|x|^2)^2}$$

$Ric(\omega) = \omega$.

we can define family ω_λ using automorphism $\varphi_\lambda : \mathbb{CP}^1 \longrightarrow \mathbb{CP}^1$ $\quad \lambda \in \mathbb{R}$

$$[z_0 : z_1] \longmapsto [z_0 : \lambda z_1]$$
$$x \longmapsto \lambda x$$

$$\omega_\lambda = \varphi_\lambda^* \omega = 2\sqrt{-1}\,\partial\bar\partial \log(1 + |\lambda x|^2) = 2\sqrt{-1}\,\partial\bar\partial \log(\lambda^{-2} + |x|^2)$$
$$= \omega + \sqrt{-1}\,\partial\bar\partial \log \frac{(\lambda^{-2} + |x|^2)^2}{(1+|x|^2)^2} = f_\lambda$$

$$\omega_\lambda = \omega + \sqrt{-1}\,\partial\bar\partial f_\lambda$$

$$\omega_\lambda = 2\sqrt{-1}\,\frac{\lambda^2\, dx \wedge d\bar{x}}{(1+|\lambda x|^2)^2} = \lambda^2 \frac{(1+|x|^2)^2}{(1+|\lambda x|^2)^2}\,\omega$$

$$= e^{c_\lambda - f_\lambda}\,\omega$$

i.e. if we let $\tilde{f}_\lambda = f_\lambda - c_\lambda$ then

$$\det(\omega_\lambda) = e^{-\tilde{f}_\lambda}\det(\omega)$$

$$\omega_\lambda = \omega + \sqrt{-1}\,\partial\bar{\partial}\tilde{f}_\lambda$$

As $\lambda \to \infty$ we have a diverging family of solutions to

$$\det(\omega + \sqrt{-1}\,\partial\bar{\partial} f) = e^{-f}\det(\omega)$$

However the metrics ω_λ are all isometric to the round metric on $\mathbb{P}^1$.

<u>Goal for rest of course</u>: understand more geometrically what can happen to family of solutions of $Ric(\omega_t) = t\omega_t + (1-t)\omega$.